

Organic Chemistry: Functional Groups

Subject: Chemistry

What Are Functional Groups?

Functional groups are specific atoms or groups of atoms within a molecule that are responsible for its characteristic chemical reactions. They largely determine the physical and chemical properties of the molecule and allow chemists to predict reactivity.

Key Functional Groups

- Alcohols (-OH): Polar, capable of hydrogen bonding, high boiling points relative to alkanes.
- Aldehydes (-CHO): Oxidizable to carboxylic acids; found at the end of carbon chains.
- Ketones (C=O): Carbonyl group flanked by two carbon atoms; not easily oxidized.
- Carboxylic Acids (-COOH): Weak acids that can form esters and amides.
- Amines (-NH₂): Act as bases, can accept protons; important in amino acids.

Key Reactions

Substitution: $\text{R-OH} + \text{HBr} \rightarrow \text{R-Br} + \text{H}_2\text{O}$

Esterification: $\text{R-COOH} + \text{R'-OH} \rightarrow \text{R-COO-R'} + \text{H}_2\text{O}$

Oxidation: $\text{R-CHO} + [\text{O}] \rightarrow \text{R-COOH}$

Exam Topics

1. Identify functional groups from structural and condensed formulas.
2. Predict the reactivity and polarity of a molecule based on its functional groups.
3. Write the mechanism for esterification.
4. Distinguish between aldehydes and ketones.
5. Explain why alcohols have higher boiling points than alkanes of similar molecular weight.